MA3236 NONLINEAR PROGRAMMING
Semester 1, AY 2026/2027
Tutorial 6

Review: Lectures 12–13

- **Smooth constrained problem and Lagrangian.** For an open domain $\mathcal{D} \subseteq \mathbf{R}^n$,

$$\mathcal{S} = \{\mathbf{x} \in \mathcal{D} : g_i(\mathbf{x}) = 0,\ h_j(\mathbf{x}) \le 0\}, \qquad J(\mathbf{x}) = \{j : h_j(\mathbf{x}) = 0\}.$$

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\mathbf{x}) + \sum_{i=1}^{m} \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^{p} \nu_j h_j(\mathbf{x}).$$

  The equality multipliers λ_i's are unrestricted; the inequality multipliers ν_j's are nonnegative.

- **KKT first-order conditions and KKT points.** A feasible point $\mathbf{x}^*$ is a *KKT point* if multipliers exist with

$$\begin{aligned} \nabla_{\mathbf{x}} L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) &= \mathbf{0} && && \text{(stationarity)},\\ g_i(\mathbf{x}^*) &= 0, && h_j(\mathbf{x}^*) \le 0 && \text{(primal feasibility)},\\ \nu_j^* &\ge 0 && && \text{(dual feasibility)},\\ \nu_j^* h_j(\mathbf{x}^*) &= 0 && && \text{(complementary slackness)}. \end{aligned}$$

  Only active inequalities can contribute to stationarity. In the force interpretation, the forces are $-\nabla f$, $-\lambda_i \nabla g_i$, and $-\nu_j \nabla h_j$.

- **LICQ and KKT first-order necessary conditions (FONC).** LICQ holds at $\mathbf{x} \in \mathcal{S}$ if the family

$$\{\nabla g_i(\mathbf{x}) : i = 1, \dots, m\} \ \cup\ \{\nabla h_j(\mathbf{x}) : j \in J(\mathbf{x})\} \quad \text{is linearly independent.}$$

  For continuously differentiable data,

$$\mathbf{x}^* \text{ local minimizer} + \text{LICQ} \implies \mathbf{x}^* \text{ is a KKT point with unique multipliers.}$$

  Warnings. A KKT point need not be a local minimizer; LICQ is sufficient, not necessary, for KKT multipliers to exist; an inactive inequality has $\nu_j^* = 0$, but an active inequality can also have $\nu_j^* = 0$ (contact without load).

Review: Lectures 12–13 (continued)

- **Tangent and linearized cones.**

$$T_{\mathcal{S}}(\mathbf{x}^*) = \left\{ \mathbf{d} : \begin{array}{l} \exists\, t^{(k)} \downarrow 0,\ \mathbf{x}^{(k)} \in \mathcal{S},\ \mathbf{x}^{(k)} \to \mathbf{x}^*, \\ (\mathbf{x}^{(k)} - \mathbf{x}^*)/t^{(k)} \to \mathbf{d} \end{array} \right\}.$$

$$T_{\text{lin}}(\mathbf{x}^*) = \{\mathbf{d} : \nabla g_i(\mathbf{x}^*)^T \mathbf{d} = 0;\ \nabla h_j(\mathbf{x}^*)^T \mathbf{d} \le 0\ (j \in J(\mathbf{x}^*))\}.$$

  Always $T_{\mathcal{S}} \subseteq T_{\text{lin}}$; LICQ gives equality. At a local minimizer, $\nabla f(\mathbf{x}^*)^T \mathbf{d} \ge 0$ for all $\mathbf{d} \in T_{\mathcal{S}}(\mathbf{x}^*)$.

- **Critical cone: first-order-neutral directions.**

$$C_{\text{crit}}(\mathbf{x}^*) = \{\mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*) : \nabla f(\mathbf{x}^*)^T \mathbf{d} = 0\}.$$

  At a KKT triple, define $J_+ = \{j \in J(\mathbf{x}^*) : \nu_j^* > 0\}$ and $J_0 = \{j \in J(\mathbf{x}^*) : \nu_j^* = 0\}$. Then

$$C_{\text{crit}}(\mathbf{x}^*) = \left\{ \mathbf{d} : \begin{array}{ll} \nabla g_i(\mathbf{x}^*)^T \mathbf{d} = 0, & i = 1, \ldots, m, \\ \nabla h_j(\mathbf{x}^*)^T \mathbf{d} = 0, & j \in J_+, \\ \nabla h_j(\mathbf{x}^*)^T \mathbf{d} \le 0, & j \in J_0 \end{array} \right\}.$$

  Loaded walls impose equality; unloaded active walls retain inequality.

- **KKT second-order conditions.** For twice continuously differentiable data, set

$$\mathbf{H}_L = \nabla^2_{\mathbf{xx}} L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*).$$

  SONC: local minimizer + LICQ $\Longrightarrow \mathbf{d}^T \mathbf{H}_L \mathbf{d} \ge 0$ for every $\mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*)$.

  SOSC: KKT first-order conditions + $\mathbf{d}^T \mathbf{H}_L \mathbf{d} > 0$ for every nonzero $\mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*)$ $\Longrightarrow$ strict local minimum (indeed, quadratic growth).

  Warnings. Use $\mathbf{H}_L$, not merely $\mathbf{H}_f$. SONC is not sufficient; SOSC is not necessary. If $C_{\text{crit}} = \{\mathbf{0}\}$, SOSC is vacuously satisfied.

- **Reduced Hessian and logical limits.** If $C_{\text{crit}} = \text{Range}(Z)$ is a subspace and Z has independent columns, SONC/SOSC reduce to $Z^T \mathbf{H}_L Z \succeq 0$ / $Z^T \mathbf{H}_L Z \succ 0$. This applies to equality-only problems and/or under strict complementarity ($J_0 = \emptyset$). A general cone is not a subspace.

1. Consider

$$\begin{array}{ll} \min & f(\mathbf{x}) = -\dfrac{(x_1-1)^2}{16} - 4x_2^2 \\ \text{s.t.} & h_1(\mathbf{x}) = -x_1 - 8x_2 - 3 \le 0, \quad h_2(\mathbf{x}) = x_1 + 8x_2 - 5 \le 0, \\ & h_3(\mathbf{x}) = -x_1 + 8x_2 - 3 \le 0, \quad h_4(\mathbf{x}) = x_1 - 8x_2 - 5 \le 0. \end{array}$$

It is given that all feasible points satisfy LICQ and that

$$\mathbf{a} = (-3, 0)^T, \qquad \mathbf{b} = (5, 0)^T, \qquad \mathbf{c} = (1, 1/2)^T$$

are global minimizers.

(a) Find the active sets and verify the KKT FONC at these three points.

(b) Find J_+, J_0, and the critical cone at each point. Explain why SOSC holds even though $\mathbf{H}_f \prec 0$ on $\mathbf{R}^2$.

(c) Keep the constraints but replace the objective by $\widetilde{f}(\mathbf{x}) = f(\mathbf{x}) + 4x_2$. At $\mathbf{a}$, verify the KKT first-order conditions, identify an active zero-multiplier constraint, and use SONC to decide whether $\mathbf{a}$ can be a local minimizer of the modified problem.

Solution. The objective gradient and Hessian are

$$\nabla f(\mathbf{x}) = (-(x_1-1)/8, -8x_2)^T, \qquad \mathbf{H}_f = \begin{pmatrix} -1/8 & 0 \\ 0 & -8 \end{pmatrix}.$$

The active sets and unique multipliers are

point	J	$(\nu_1, \nu_2, \nu_3, \nu_4)$	J_0
$\mathbf{a}$	$\{1, 3\}$	$(1/4, 0, 1/4, 0)$	$\emptyset$
$\mathbf{b}$	$\{2, 4\}$	$(0, 1/4, 0, 1/4)$	$\emptyset$
$\mathbf{c}$	$\{2, 3\}$	$(0, 1/4, 1/4, 0)$	$\emptyset$

For example, at $\mathbf{a}$,

$$(1/2, 0)^T + \tfrac{1}{4}(-1, -8)^T + \tfrac{1}{4}(-1, 8)^T = \mathbf{0}.$$

The other two balances are

$$(-1/2, 0)^T + \tfrac{1}{4}(1, 8)^T + \tfrac{1}{4}(1, -8)^T = \mathbf{0},$$

$$(0, -4)^T + \tfrac{1}{4}(1, 8)^T + \tfrac{1}{4}(-1, 8)^T = \mathbf{0}.$$

All listed multipliers are nonnegative and inactive multipliers vanish, so all four parts of the KKT first-order conditions hold.

At each point, $J_+ = J$. The two loaded active gradients are independent; requiring orthogonality to both gives $C_{\text{crit}} = \{\mathbf{0}\}$. There is no nonzero critical

direction on which SOSC must be tested, so SOSC holds and each point is a strict local minimizer. Positive definiteness of the ambient Hessian is not required. These points are not strict *global* minimizers because they have the same objective value.

For the modified objective, $\nabla \widetilde{f}(\mathbf{a}) = (1/2, 4)^T$, and stationarity gives

$$(\nu_1, \nu_2, \nu_3, \nu_4) = (1/2, 0, 0, 0).$$

Thus $J_+ = \{1\}$, $J_0 = \{3\}$, and

$$C_{\text{crit}}(\mathbf{a}) = \{\mathbf{d} : -d_1 - 8d_2 = 0,\ -d_1 + 8d_2 \le 0\} = \{t(8, -1)^T : t \ge 0\}.$$

The constraints are affine, so $\mathbf{H}_L = \mathbf{H}_{\widetilde{f}} = \mathbf{H}_f$. The nonzero critical direction $(8, -1)^T$ gives

$$(8, -1)\mathbf{H}_L(8, -1)^T = -16 < 0.$$

SONC fails. Because LICQ holds, $\mathbf{a}$ is not a local minimizer. The active constraint h_3 has zero multiplier, so replacing its critical-cone inequality by an equality would incorrectly remove this rejecting direction.

2. It is given that $\mathbf{x}^* = (4, 0)^T$ is a global maximizer of

$$\begin{array}{ll} \max & 2x_1 + x_2 \\ \text{s.t.} & h_1(\mathbf{x}) = -x_1 + x_2 \le 0, \\ & h_2(\mathbf{x}) = (x_1 - 4)^3 + 4x_2 \le 0, \\ & h_3(\mathbf{x}) = -x_1 \le 0, \qquad h_4(\mathbf{x}) = -x_2 \le 0. \end{array}$$

   (a) Rewrite the objective as a minimization. Do the KKT first-order conditions hold at $\mathbf{x}^*$? Explain why your answer does not contradict KKT FONC.

   (b) Compute $T_{\text{lin}}(\mathbf{x}^*)$ and $T_{\mathcal{S}}(\mathbf{x}^*)$. Verify the tangent-cone first-order necessary condition and identify a direction that is linearly feasible but is not tangent.

Solution. Use $F(\mathbf{x}) = -2x_1 - x_2$. The active set is $J = \{2, 4\}$, with

$$\nabla h_2(\mathbf{x}^*) = (0, 4)^T, \qquad \nabla h_4(\mathbf{x}^*) = (0, -1)^T.$$

Stationarity would require

$$(-2, -1)^T + \nu_2(0, 4)^T + \nu_4(0, -1)^T = \mathbf{0}, \qquad \nu_2, \nu_4 \ge 0,$$

whose first component is impossible. LICQ fails because the active gradients are dependent; KKT FONC therefore do not guarantee multipliers.

Linearization gives $4d_2 \le 0$ and $-d_2 \le 0$, so

$$T_{\text{lin}}(\mathbf{x}^*) = \{(d_1, 0)^T : d_1 \in \mathbf{R}\}.$$

For a tangent sequence with scale $t^{(k)}$, the first component satisfies $(x_1^{(k)} - 4)/t^{(k)} \to d_1 \le 0$. In particular, $4 - x_1^{(k)} = O(t^{(k)})$, and the hint gives $x_2^{(k)}/t^{(k)} = O((t^{(k)})^2) \to 0$. Conversely, every $d_1 \le 0$ is realized by the feasible path $(4 + td_1, 0)$ for sufficiently small $t > 0$. Hence

$$T_{\mathcal{S}}(\mathbf{x}^*) = \{(d_1, 0)^T : d_1 \le 0\}.$$

For every such direction, $\nabla F(\mathbf{x}^*)^T \mathbf{d} = -2d_1 \ge 0$, as required. The fictitious direction $(1, 0)^T$ belongs to T_{lin} but not to $T_{\mathcal{S}}$ and has negative inner product with ∇F. Thus the cone mismatch explains the failure of KKT multipliers.

3. Consider

$$\begin{array}{ll} \min & f(\mathbf{x}) = -x_3^2 \\ \text{s.t.} & h_1(\mathbf{x}) = x_1^2 + x_2^2 - x_3 \le 0, \\ & h_2(\mathbf{x}) = x_1^2 + x_2^2 + x_3^2 - 1 \le 0. \end{array}$$

   (a) Prove that a global minimizer exists, that LICQ holds at every feasible point, and find all global minimizers.

   (b) At the minimizer, find the KKT multipliers and the critical cone.

   (c) Compute $\mathbf{H}_L$ and verify SOSC. Explain why the singularity of the ambient $\mathbf{H}_L$ does not prevent this certificate.

Solution. The feasible set is nonempty (it contains $\mathbf{0}$), closed, and contained in the unit ball. It is compact; continuity gives a global minimizer.

If only h_1 is active, its gradient $(2x_1, 2x_2, -1)^T$ is nonzero. If only h_2 is active, its gradient $2\mathbf{x}$ is nonzero on the sphere. If both are active, then

$$x_3 = x_1^2 + x_2^2, \qquad x_3 + x_3^2 = 1.$$

Thus $x_3 = (\sqrt{5} - 1)/2 > 0$ and $(x_1, x_2) \ne (0, 0)$. If a combination of the two active gradients is zero, their first two components force the coefficients to be opposites. The third component then forces both to vanish because $1 + 2x_3 > 0$. So the gradients are independent. With no active constraints, LICQ holds vacuously.

Feasibility gives $0 \le x_3 \le 1$, whence $f(\mathbf{x}) \ge -1$. Equality forces $x_3 = 1$ and, by the sphere inequality, $x_1 = x_2 = 0$. The unique global minimizer is therefore

$$\mathbf{x}^* = (0, 0, 1)^T, \qquad f(\mathbf{x}^*) = -1.$$

Here $J = \{2\}$, and the KKT first-order conditions give $\nu_1^* = 0$, $\nu_2^* = 1$. Since the only active multiplier is positive,

$$C_{\text{crit}}(\mathbf{x}^*) = \{\mathbf{d} : d_3 = 0\}.$$

The Lagrangian Hessian is

$$\mathbf{H}_L = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix} + 2I = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

For a nonzero critical direction, $\mathbf{d}^T \mathbf{H}_L \mathbf{d} = 2(d_1^2 + d_2^2) > 0$. Thus SOSC holds. The zero-eigenvalue direction is normal to the critical subspace and is irrelevant to this test. For $Z = [(1,0,0)^T \ (0,1,0)^T]$, the reduced Hessian is $2I_2 \succ 0$.

4. Consider

$$\begin{array}{ll} \min & f(\mathbf{x}) = x_1 - x_2 \\ \text{s.t.} & g_1(\mathbf{x}) = x_1^2 + x_2^2 - 2 = 0, \\ & g_2(\mathbf{x}) = x_2 - x_3^3 = 0. \end{array}$$

   (a) Find $\alpha > 0$ with $|x_i| \le \alpha$ for every feasible point and $i = 1, 2, 3$. Explain why a global minimizer exists.

   (b) Prove LICQ at every feasible point and find all global minimizers.

   (c) Find all KKT points and their multipliers. Use critical cones and reduced Hessians to classify the points whenever second-order information gives a conclusion.

   (d) For the remaining points, use the local feasible parametrization

$$\mathbf{x}_\pm(t) = (\pm\sqrt{2 - t^6}, t^3, t)^T$$

   to decide whether they are local minima, local maxima, or saddle points relative to the feasible set.

Solution. The equalities give $|x_1|, |x_2| \le \sqrt{2}$ and $|x_3| \le 2^{1/6} \le \sqrt{2}$, so $\alpha = \sqrt{2}$ suffices. The feasible set is nonempty and compact; a global minimizer exists.

The constraint gradients are

$$\nabla g_1 = (2x_1, 2x_2, 0)^T, \qquad \nabla g_2 = (0, 1, -3x_3^2)^T.$$

The second gradient is always nonzero. If $x_1 \neq 0$, the gradients are independent by their first components. If $x_1 = 0$, then $x_2 = \pm\sqrt{2}$ and $x_3 \neq 0$, so their third components show independence. Thus LICQ holds everywhere.

On $x_1^2 + x_2^2 = 2$, Cauchy–Schwarz gives $x_1 - x_2 \ge -2$, with equality only at $(x_1, x_2) = (-1, 1)$. The second equality then gives $x_3 = 1$. The unique global minimizer is $(-1, 1, 1)^T$.

The KKT first-order equations are

$$1 + 2\lambda_1 x_1 = 0, \qquad -1 + 2\lambda_1 x_2 + \lambda_2 = 0, \qquad -3\lambda_2 x_3^2 = 0, \qquad g_1 = g_2 = 0.$$

If $x_3 \neq 0$, then $\lambda_2 = 0$, and the first two equations give $x_2 = -x_1 = \pm 1$. If $x_3 = 0$, feasibility gives $x_2 = 0$ and $x_1 = \pm\sqrt{2}$, while stationarity gives $\lambda_2 = 1$. Thus all KKT points are

$\mathbf{x}$	(λ_1, λ_2)
$(-1, 1, 1)^T$	$(1/2, 0)$
$(1, -1, -1)^T$	$(-1/2, 0)$
$(\sqrt{2}, 0, 0)^T$	$(-1/(2\sqrt{2}), 1)$
$(-\sqrt{2}, 0, 0)^T$	$(1/(2\sqrt{2}), 1)$

There are only equalities, so $C_{\text{crit}} = \{\mathbf{d} : \nabla g_1^T \mathbf{d} = \nabla g_2^T \mathbf{d} = 0\}$. Furthermore,

$$\mathbf{H}_L = \text{diag}(2\lambda_1, 2\lambda_1, -6\lambda_2 x_3).$$

At the first two points, the critical subspace has basis $Z = (3, 3, 1)^T$. The reduced Hessian is 18 at $(-1, 1, 1)^T$ and -18 at $(1, -1, -1)^T$. SOSC gives a strict local minimum at the former. Applying SOSC to the objective $-f$ gives a strict local maximum at the latter (it is also the unique global maximum).

At $(\pm\sqrt{2}, 0, 0)^T$, a critical-subspace basis is $Z = (0, 0, 1)^T$ and $Z^T \mathbf{H}_L Z = 0$. SONC holds, but SOSC fails; no classification follows from these second-order tests alone. Along the given feasible parametrizations,

$$f(\mathbf{x}_\pm(t)) - f(\mathbf{x}_\pm(0)) = -t^3 + O(t^6).$$

This has opposite signs for sufficiently small positive and negative t. Both remaining points are saddle points on the feasible set. They illustrate why passing SONC is not sufficient for minimality.